25901103 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Core Course (DSC) - MG4DSCBTS201 - CELL BIOLOGY

(2024 ADMISSION ONWARDS)

Duration: 1 hour 30 minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Fill in the blanks

Answer all questions

Each question carries **1** marks

1. The main fuction of chloroplast is to produce…….....through photosynthesis. [K] / [CO1]
2. ….....transport requires energy to move molecules against a concentration gradient. [K] / [CO2]
3. Formation of shoot meristems in tissue culture is a _____-mediated response. [U] / [CO3]
4. ….....are membrane bound organelles containing digestive enzymes. [K] / [CO1]
5. ….....are the protective caps at the end of chromosome. [K] / [CO2]
6. The region where spindle fibers attach is called the _____. [U] / [CO2]
7. The outermost layer of a plant cell is ……..... [K] / [CO1]
8. ….....molecules regulate membrane fluidity. [K] / [CO2]
9. The three subunits of G-protein are _____, _____ and _____. [K] / [CO3]
10. The process of generating energy in mitochondria is called…..... [K] / [CO1]

[1x10 = 10]

Part B

Short Answer Questions

Answer any **10** questions

Each question carries **2** marks

11. Which cytoskeletal elements are associated with desmosomes?	[K] / [CO2]
12. Differentiate between karyokinesis and cytokinesis.	[U] / [CO3]
13. What is the role of intermediate filaments?	[U] / [CO1]
14. What are selectins? Mention one of their functions.	[U] / [CO2]
15. What are stem cells?	[K] / [CO3]
16. What is the role of histone proteins in nucleosomes?	[K] / [CO2]
17. Who discovered cells and in which year?	[K] / [CO1]
18. What is meant by “critical mass” in quorum sensing?	[U] / [CO3]
19. Comment on NOR	[K] / [CO1]
20. Name any two CDK inhibitors.	[U] / [CO3]
21. Define prokaryotic cell and give one example.	[K] / [CO1]
22. Mention any two differences between apoptosis and necrosis.	[K] / [CO3]

[2x10 = 20]

Part C

Short Essay Questions

Answer any **5** questions

Each question carries **4** marks

23. Explain the different forms of chemical signaling in multicellular organisms.	[U] / [CO3]
24. Explain the role of ribosomes in protein synthesis.	[K] / [CO1]
25. Describe the functions of SER.	[K] / [CO2]
26. Describe the structure and formation of RER .Write two functions also.	[K] / [CO2]
27. Describe the fuctions of vacuoles in plant cells.	[K] / [CO1]
28. Explain the significance of crossing over during meiosis.	[K] / [CO2]
29. Describe the functions of golgi complex.	[K] / [CO1]

[4x5 = 20]